FIITJEE

JEE (Advanced)-2018

ALL INDIA TEST SERIES

FULL TEST – X

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 231**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-C & Section-D**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. ***Section-A (01 – 03, 24 – 26, 47 – 49)*** *contains 9 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*
   Section-A (04 – 08, 27 – 31, 50 – 54) *contains 15 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer.*
   Partial Marks **+1** for each correct option provided no incorrect options is selected.
   Section-A (09 – 10, 32 – 33, 55 – 56) *contains 3 paragraphs. Based upon paragraph, 2 multiple choice questions have to be answered. Each question has* ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer. There is no negative marking.*
2. ***Section-C (11 – 20, 34 – 43, 57 – 66)*** *contains 30 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*
3. ***Section-D (21 – 23, 44 – 46, 67 – 69)*** *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

PART – I (Physics), PART – II (Chemistry), PART – III (Mathematics):
(SECTION – D)

For questions **21 to 23, 44 to 46, 67 to 69.**
Numerical answer type questions with answer XXXXX. XX

If answer is 348.4 / 251.37 / 213

Correct Method :

0	0	3	4	8	.	4	0
0	0	2	5	1	.	3	7
0	0	2	1	3	.	0	0

Wrong Method :

	3	4	8		.	4	
3	4	8			.		4
		3	4	8	.		4
	3		4	8	.	4	
	2		5	1	.	3	7
		2	1	3	.		
		2	1	3	.	0	
		2	1	3	.		0
		3	4	8	.	4	0
		2	5	1	.	3	7
		2	1	3	.	0	0

Physics

PART – I

SECTION – A

Straight Objective Type

This section contains **3 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

1. The escape velocity at the poles of a planet is v_e. If the effect of rotation of a planet makes the weight of a body at the equator 10% less than the weight at pole, the peripheral velocity of a point on the equator of the planet is (assume the planet to be a uniform solid sphere)

(A) $\frac{v_e}{\sqrt{5}}$ (B) $\frac{v_e}{2\sqrt{5}}$

(C) $\frac{\sqrt{5}v_e}{2}$ (D) $\frac{2v_e}{5}$

2. Current I flowing along edges of one face of a cube as shown in figure-I, produces magnetic field $\vec{B} = B_0\hat{k}$ at the centre of the cube. Consider another identical cube, where current I flows along the path as shown in the figure-II. What magnetic field is produced at the centre of the second cube?

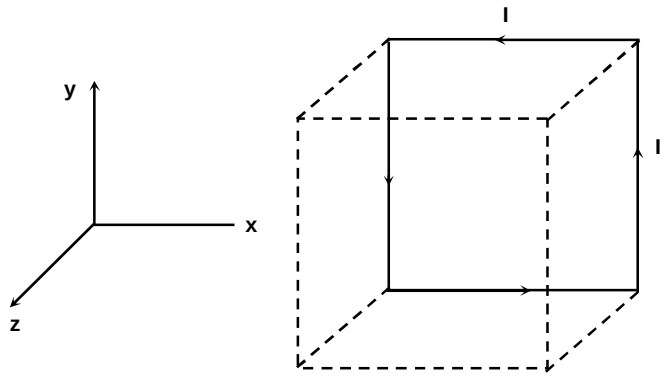


Figure – (i)

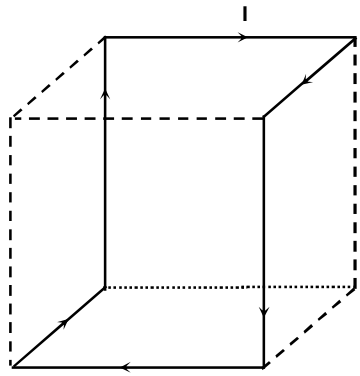


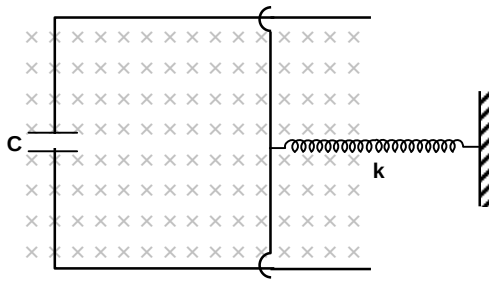
Figure – (ii)

(A) Zero (B) $B_0\left(-\hat{j}+\hat{k}+\hat{i}\right)$

(C) $B_0\left(-\hat{j}-\hat{k}+\hat{i}\right)$ (D) $B_0\left(\hat{j}+\hat{k}-\hat{i}\right)$

Space for Rough work

3. One end of a non-conducting spring of force constant k is connected to the mid-point of a conducting rod of mass m and length ℓ, which can slide freely on a pair of parallel and smooth horizontal conducting rails. The other end of the spring is connected to a rigid wall. There exists a uniform magnetic field B normal to the plane of paper and into it. Electrical resistance of the rails and rod are negligible. The rails are connected to a capacitor of capacitance C. If the wire is pushed towards left and released, the time period of its oscillation will be $\left(\text{Given } m = 2CB^2\ell^2\right)$.



(A) $T = 2\pi\sqrt{\frac{m}{k}}$

(B) $T = \pi\left(\sqrt{\frac{3m}{2k}} + \sqrt{\frac{m}{2k}}\right)$

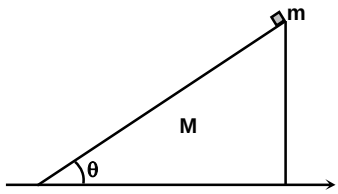
(C) $T = 2\pi\sqrt{\frac{3m}{2k}}$

(D) $T = \pi\left(\sqrt{\frac{3m}{2k}} + \sqrt{\frac{m}{k}}\right)$

Multiple Correct Choice Type

This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

4. In the figure shown, the friction between the wedge and the horizontal surface is sufficient to prevent the wedge from sliding. The mass of the wedge is M and its inclination is θ. A particle of mass m is held motionless on the top of the wedge. Friction coefficient between particle and wedge is μ. Now the particle and the wedge are released from rest. Then



(A) If $\mu > \tan\theta$, wedge will not topple for any value of m.

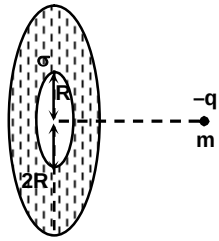
(B) If $\mu > \tan\theta$, wedge will topple for $m > \frac{M}{3\sin\theta(\sin\theta + \mu\cos\theta)}$

(C) If $\mu < \tan\theta$, wedge will topple for $m > \frac{M}{3\sin\theta(\sin\theta - \mu\cos\theta)}$

(D) If $\mu < \tan\theta$, wedge will topple for $m > \frac{M}{6\sin\theta(\sin\theta - \mu\cos\theta)}$

Space for Rough work

5. A charged fixed annular disc of uniform positive charge density σ has inner and outer radii R and 2R respectively. A negatively charged particle of mass m and charge – q is released on the axis of the disc from a distance x from the centre. Then



(A) The speed of the particle when reaches the centre of ring, $v = \sqrt{\frac{q\sigma}{m\varepsilon_0}\left(R + \sqrt{R^2 + x^2} - \sqrt{4R^2 + x^2}\right)}$

(B) The speed of the particle when it reaches the centre of ring, $v = \sqrt{\frac{q\sigma}{m\varepsilon_0}\left(R + \sqrt{4R^2 + x^2} - \sqrt{R^2 + x^2}\right)}$

(C) Work done by electrostatic force till the particle reaches the centre of ring $= \frac{q\sigma}{2\varepsilon_0}\left(R + \sqrt{R^2 + x^2} - \sqrt{4R^2 + x^2}\right)$

(D) If x << R, the frequency of oscillation of the particle is $f = \frac{1}{2\pi}\sqrt{\frac{q\sigma}{4\varepsilon_0 mR}}$

6. An artificial satellite of mass m revolves in a circular orbit of radius 4R around the centre of the moon (where R is the radius of the moon). Suddenly it starts experiencing a slight resistance due to cosmic dust. The resistance force depends on the speed of satellite as F = kv where k is a constant. If the radius of the moon is R and acceleration due to gravity on the moon's surface is g, then (assume that at every moment the satellite follows a circular path)

(A) The satellite will hit the moon's surface after time $t = \frac{m}{k}\ln 2$

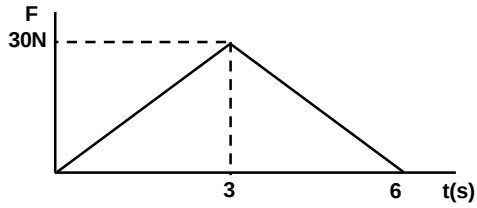
(B) The satellite will hit the moon's surface after time $t = \frac{2m}{k}\ln 2$

(C) The work done by the resistance force on the satellite till it hits the moon surface $= -\frac{3}{8}mgR$.

(D) Work done by gravitational force till the satellite hits the moon's surface = $\frac{3}{4}mgR$.

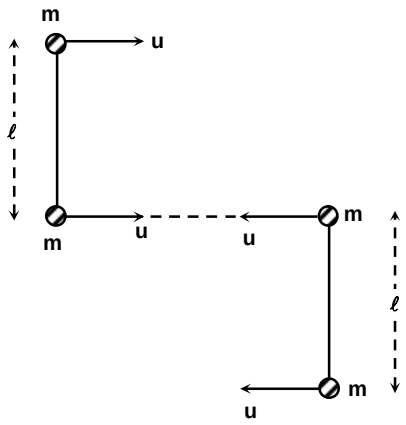
Space for Rough work

7. A block of mass m = 2 kg rests on a rough horizontal surface. The coefficient of friction between block and surface is μ = 0.5. The block is acted upon by a horizontal time varying force as shown. Then



(A) The block starts moving at time t = 1s
(B) The maximum velocity acquired by the block is 25 m/s
(C) The maximum velocity attained by the block is 20 m/s
(D) Velocity of block at t = 6 sec is 12.5 m/s

8. Two identical dumbbells are formed by welding two small identical particles each of mass m at the ends of massless rod each of length ℓ. The dumbbells are moving in free space with equal speed u without rotation towards each other as shown in the figure. Collision between the particles are elastic. Then



(A) Angular velocity of each dumbbell just after first collision is $\frac{u}{\ell}$
(B) Angular velocity of each dumbbell just after first collision is $\frac{2u}{\ell}$
(C) Speed of centre of mass of any one of the dumbbell just after collision is zero.
(D) Speed of centre of mass of any one of the dumbbell just after collision is u.

Space for Rough work

Comprehension Type

This section contains **1 group of questions.** The group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct

Paragraph for Question Nos. 9 and 10

A pendulum is suspended from a fixed point. The mass of the bob of the pendulum is m and the length of string is ℓ. It is initially at rest at its lowest position. A particle of mass $\frac{m}{2}$ moving horizontally along negative x-direction with velocity $\sqrt{2g\ell}$ collides with the bob and comes to rest. When the bob comes to rest, another particle Q of mass m moving horizontally along y-direction collides with the bob and sticks to it. It is observed that the bob now moves along a horizontal circle. There is horizontal floor 2ℓ distance below the point of suspension of the pendulum.

9. Which of the following is possible value of speed of particle Q just before its collision with the bob?

(A) $\sqrt{\frac{7\ell g}{3}}$ (B) $\sqrt{\frac{3\ell g}{4}}$

(C) $\sqrt{\frac{7\ell g}{4}}$ (D) $\sqrt{\frac{4\ell g}{3}}$

10. If string breaks during the circular motion, the bob will hit the floor after a time equal to

(A) $\sqrt{\frac{4\ell}{g}}$ (B) $\sqrt{\frac{4\ell}{2g}}$

(C) $\sqrt{\frac{5\ell}{2g}}$ (D) $\sqrt{\frac{\ell}{2g}}$

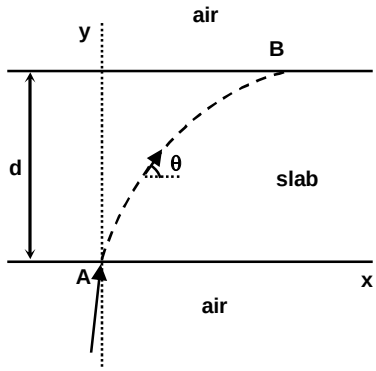
Space for Rough work

SECTION – C

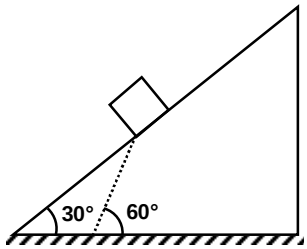
(One Integer Value Correct Type)

This section contains **10 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

11. A ray of light travelling in air is incident at grazing angle (incident angle $\approx 90°$) on a long rectangular slab of a transparent medium of thickness d. The point of incidence is the origin A (0, 0). The medium has a variable index of refraction $\mu = \sqrt{1+e^{x/d}}$. Determine the x coordinate (in meter) of the point B, where the ray meets the upper surface of slab air boundary (Take d = (25/7) m and ln2 = 0.7).



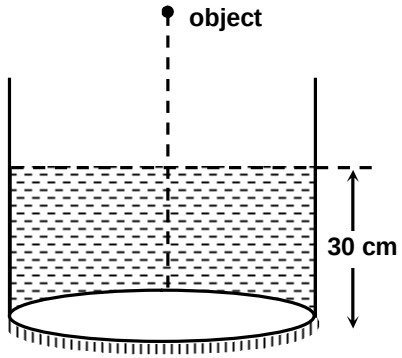
12. A wedge of mass 3 kg is lying at rest on a frictionless horizontal surface. A block of mass m is released from rest on the wedge. If the path of block (with respect to the ground) makes an angle 60° with horizontal after it is released, the mass of block (in kg) is



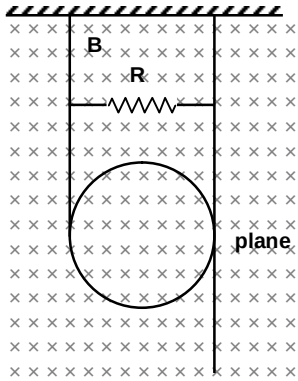
13. A solid object of uniform cross section area and relative density 0.2 is pressed down and held at rest inside water surface so that its upper surface is in level with the water surface. Length of the object is $\ell = 1.2\text{m}$. When released the object jumps out of water moving vertically. The velocity of the object (in m/s) when it just comes out completely from the water is (Take g = 10 m /s^2).

Space for Rough work

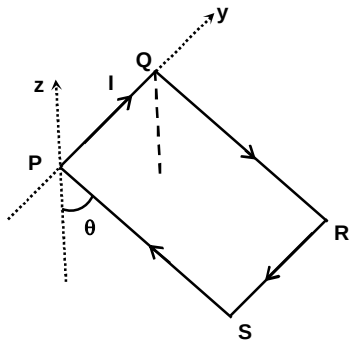
14. The bottom of a container is made of an equi-convex lens. The focal length of the lens in air is 60 cm. Now its lower surface is silvered. The refractive index of the material of the lens is 1.5. Water is filled in the container to a depth 30 cm. When an object is placed at a distance $\frac{27}{n}$ cm above the water surface on the principle axis of lens, the final image of the object coincides with the object. Find the value of n. (given $\mu_{water} = 4/3$)



15. A conducting light string is wound on the rim of a metal ring of radius r and mass m. The free end of the string is fixed to the ceiling. A vertical infinite conducting smooth plane is always tangent to the ring. The ring is released and simultaneously a uniform magnetic field of strength B perpendicular to the plane of the ring is switched on. The string and the plane are connected by a resistance R. The ring is always immersed in the magnetic field. If the terminal velocity of the ring is $\frac{mgR}{nB^2r^2}$, find the value of n.



16. A square current carrying loop of side length ℓ and carrying a current I is hinged about its horizontal side PQ (Parallel to y axis). There is a uniform magnetic field $\vec{B} = B_0\left(\hat{i} - \hat{k}\right)$ in the region. The linear mass density of loop is λ and acceleration due to gravity is g. The loop remains in equilibrium when it's plane makes an angle $\theta = 37°$ with the vertical (z-axis). If the value of B_0 is $\frac{n\lambda g}{7\,I}$, find the value of n. (given sin 37°= 3/5)



Space for Rough work

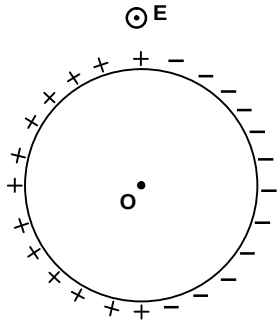
17. When the potential difference applied to an x-ray tube is 10 kV the minimum wavelength of continuous x-ray spectrum is λ_0. Further when the potential difference applied is increased from 10 kV to 20 kV the difference of wavelengths of K_α line and minimum wavelength of continuous x-ray increases to twice of its initial value. Atomic number of anode material of x-ray tube is z. If the value of z – 1 equals $\sqrt{\frac{n}{9\lambda_0 R}}$, then find the value of n.

18. The radiation emitted when an electron jumps from n = 4 to n = 3 in a Li^{++} ion (atomic no. z = 3) falls on a metal surface to produce photoelectron. When the photoelectron with maximum kinetic energy are made to move perpendicular to a uniform magnetic field of 2×10^{-4} T, it follows a circular path of radius $\sqrt{\frac{9.1}{1.6}}$ cm. The work function of metal (in eV) to the nearest integer is

(given mass of electron m = 9.1×10^{-31} kg and charge on electron = -1.6×10^{-19} C)

19. A car begins to move at time t = 0 and then accelerates along a straight horizontal track with a velocity given by $v = 2t^2$ for $0 \le t \le 2$, where t is time in second. After the end of acceleration, the car continues to move at constant speed. A small block initially at rest on the floor of the car begins to slip at t = 1 sec and stops sliding at t = 3 sec. If the coefficient of static and kinetic friction between the block and the floor are μ_S and μ_K respectively, find the value of $\frac{3\mu_S}{\mu_K}$.

20. A uniform ring of mass m and radius R is placed freely in a gravity free region. Half of the ring is positively charged with uniform linear charge density $+\lambda$ and other half is negatively charged with uniform charged density $-\lambda$. A uniform electric field E is switched on along the normal to the plane of the ring. If the angular speed of the ring when it turns through an angle 90° is $n\left(\sqrt{\frac{\lambda E}{m}}\right)$, find the value of n.



Space for Rough work

SECTION – D

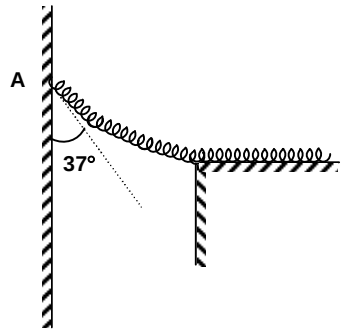
(Numerical Answer Type)

This section contains **3 questions**. Each question, when worked out will result in **numerical answer type** (XXXXX.XX)

21. In a hydrogen like atom, an electron is revolving in an orbit having quantum number n. Its frequency of revolution is found to be 13.2×10^{15} Hz. Energy required to completely remove this electron to infinity from the above orbit of atom is 54.4 eV. In a time interval of 7 nanosecond the electron jumps back to orbit having quantum number n/2. If the average torque acting on the electron during the above process is τ Nm, then find the value of $\tau \times 10^{26}$.
(given $\frac{h}{\pi} = 2.1\times10^{-34}\,J-s$, frequency of revolution of electron in the ground state of H-atom = 6.6×10^{15} Hz)

22. A screw gauge with a pitch of 0.5 mm and a circular scale with 50 divisions is used to measure the diameter of a uniform wire. Before starting the measurement, it is found that when the two jaws of the screw gauge are brought in contact, the 47th division of the circular scale coincides with the main scale line and that the zero of the main scale is barely visible. When the diameter of wire is measured by this screw gauge the main scale reads 1.10 cm and the 12th division of the circular scale coincides with the main scale line. Find the diameter (in mm) of the cylinder.

23. A uniform chain of length L has one of its end attached to a vertical wall at point A, while $\frac{3L}{4}$ of the length of the chain is lying on table as shown in figure. Tangent to the chain at A makes an angle 37° with the wall. Find the minimum coefficient of friction between table and chain so that chain remains in equilibrium.



Space for Rough work

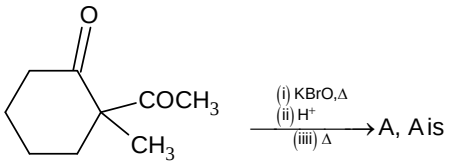
Chemistry PART – II

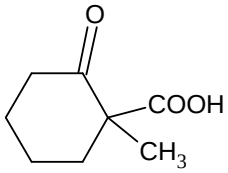
SECTION – A

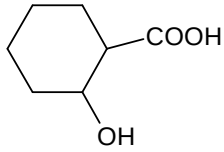
Straight Objective Type

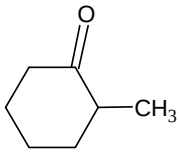
This section contains **3 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

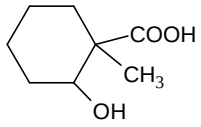
24. $KMnO_4$ in neutral/weakly alkaline medium oxidizes thiosulphate to:
(A) SO_3^{2-} (B) $S_2O_4^{2-}$
(C) SO_4^{2-} (D) $S_4O_6^{2-}$

25. 

(A) 

(B) 

(C) 

(D) 

26. A red hot sample of atomic hydrogen in a tube was subjected to electrical discharge. The approximate voltage at which the first indication of current flow was observed:
(A) 13.6 V (B) 10.2 V
(C) 1.51 V (D) 1.89 V

Space for Rough work

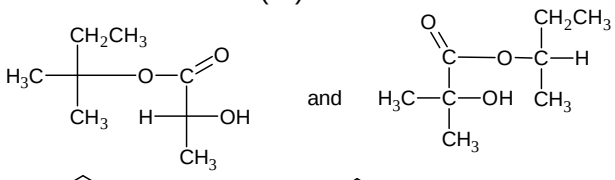
Multiple Correct Choice Type

This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

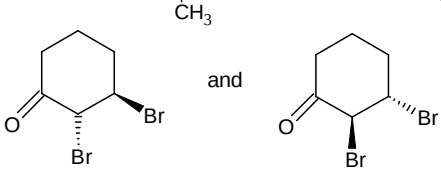
27. Which of the following pairs of compound could be separated by recrystallization?

(A) meso-tarteric acid and ($\pm$)-tartaric acid.

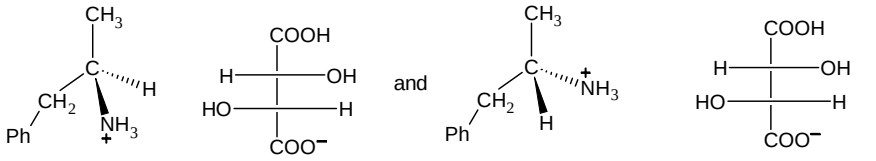
(B)



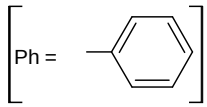
(C)



(D)



(an acid-base salt)



28. Select reaction/s in which anion of main ionic product do not have σ-bond.

(A) $Li + O_2$ (excess) $\rightarrow$ (B) Na (excess) $+ O_2 \rightarrow$

(C) $Ca + F_2$ (excess) $\rightarrow$ (D) $NaOH$ (cold) $+ Cl_2 \rightarrow$

Space for Rough work

29. In the following conversion

$$D-\text{glucopyranose} \xrightarrow[EtONa]{MeNO_2} ? \xrightarrow{H_2SO_4} ? \xrightarrow{H_2/Ni} \text{Product(s)}$$

Which of the following is/are correct about final product(s)?

(A) optically active
(B) optically inactive
(C) optically active as well as optically inactive
(D) all are correct

30. Which of the following is **incorrect** statement for an ideal gas?

(A) When temperature increases at constant pressure, collision frequency increases.
(B) Mean free path decreases by increasing temperature at constant pressure.
(C) If all gas molecules are assumed to be rigid sphere of negligible volume, the only possible molecular motion is translation.
(D) Kinetic energy decreases by increasing pressure at constant temperature.

31. There are process which are used to determine some important properties of colloidal solution. By which of the following processes, the nature of electrical charge on colloidal particles can be determined?

(A) ultramicroscopic
(B) electrophoresis
(C) dialysis
(D) electroosmosis

Space for Rough work

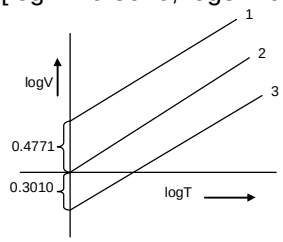
Comprehension Type

This section contains **1 group of questions.** The group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct

Paragraph for Question Nos. 32 to 33

When a sample of an ideal gas is changed from an initial state to a final state. Various curves can be plotted for the process like P – V curve, V – T curve, P – T curve etc. For example P – V curve for a fixed amount of an ideal gas at constant temperature is a rectangular, V – T curve for a fixed amount of an ideal gas at constant pressure is straight line and P – T curve for a fixed amount of an ideal gas is again straight line. However, the shapes may vary if the constant parameters are also changed.

32. At constant P of 0.0821 atm, logV v/s log T graph is plotted for 3 samples of ideal gas as shown. Value of number of moles of gas in these three samples is respectively:
[log2 = 0.3010, log3 = 0.4771]



(A) $3,1,\frac{1}{2}$ (B) $\frac{1}{2},1,3$

(C) $1,\frac{1}{2},3$ (D) $\frac{1}{2},3,1$

33. Two moles of an ideal gas is changed from its initial state (16 atm, 6L) to final state (4 atm, 15 L) in such a way that this change can be represented by a straight line in P – V curve. The maximum temperature attained by the gas during the above change is: (take $R = \frac{1}{12} L\ atm\ K^{-1}\ mol^{-1}$)

(A) 324 K (B) 648 K

(C) 1296 K (D) 972 K

Space for Rough work

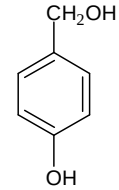
SECTION – C

(One Integer Value Correct Type)

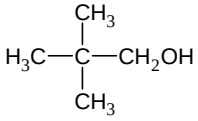
This section contains **10 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

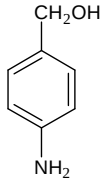
34. Compound(x) (C_8H_7N) on treatment with cold $SnCl_2$/dil.HCl (aqueous), produces y(C_8H_8O). y on heating with sodium acetate in acetic anhydride followed by acidification forms (z) ($C_{10}H_{10}O_2$). All these compounds (x, y, z) on oxidation under severe conditions form a diacid which gives only one mononitro isomer in nitrating mixture, how many products will be formed on reductive ozonlysis of (z)?

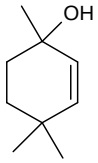
35. How many of the following can give aldehyde by the oxidation with Jones reagent?

(a) 

(b) $CH_2 = CH - CH_2OH$

(c) 

(d) 

(e) 

36. From the following the complex that is expected to show optical isomerism is/are:
(1) $[Ni(en)_3]^{2+}$ (2) $[Fe(ox)Cl_4]^{3-}$
(3) $[Cr(NH_3)_3Cl_2(NO_3)]$ (4) $[Zn(NO_3)_2(NH_3)(CO)]$

37. At 263 K a solid compound $A(H_2O)_8$ show following equilibria (Assume 263 K is sublimation point of $H_2O(s)$ under given condition):

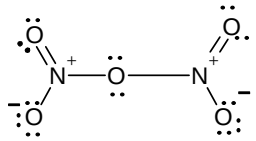
$A(H_2O)_8(s) \rightleftharpoons A(g) + 8H_2O(g)$; K_{p_1}

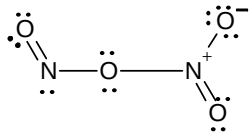
$A(H_2O)_8(s) \rightleftharpoons A(g) + 8H_2O(s)$; K_{p_2}

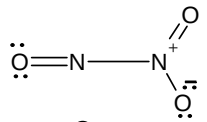
If equilibrium partial pressures of two gases A and H_2O are 0.2 bar and 0.001 bar respectively. And the vapour pressure of ice is 10^{-x} bar. What is value of 'x'?

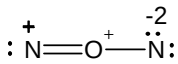
Space for Rough work

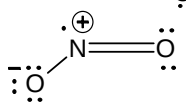
38. How many of the following is/are correct structure(s) for any oxide of nitrogen?

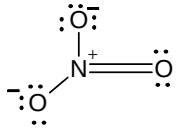
(I) 

(II) 

(III) 

(IV) 

(V) 

(VI) 

39. In 2nd order reaction: $2A \longrightarrow A_2$, the rate of formation of A_2 is 10^{-5} M sec^{-1} at 0.01 M concentration of A, and the rate of disappearance of A is $x \times 10^{-1}$, find out the value of x?

40. How many of the following reaction(s) is/are neither stereospecific nor regiospecific?

(i) $\text{trans}-2-\text{pentene} \xrightarrow{HCl}$

(ii) $1-\text{pentene} \xrightarrow[NaCl(Excess)]{Br_2}$

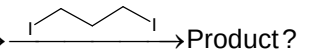
(iii) $\text{Cyclobutene} \xrightarrow[Br_2]{H_2O}$

(iv) $1-\text{ethylcyclopropene} \xrightarrow{D_2/Pt}$

(v) $\text{trans}-2-\text{pentene} \xrightarrow[NaBH_4/H_2O]{Hg(OAc)_2}$

(vi) $\text{trans}-2-\text{pentene} \xrightarrow{H^+/H_2O}$

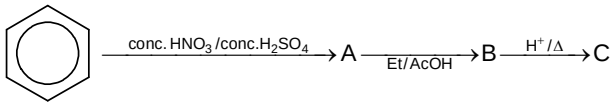
(vii) $\text{trans}-2-\text{pentene} \xrightarrow{Br_2/CCl_4}$,

41. $CH_3OH \xrightarrow[(Excess)]{NaH}$  $\longrightarrow$ Product?

How many of 'C' atoms are there in one molecule of the product?

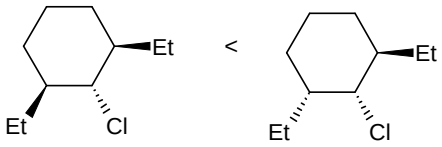
Space for Rough work

42. How many 'H' atoms are there in the final product of the following reaction sequence?

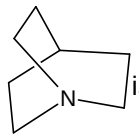


43. How many of the following option(s) is/are incorrect?

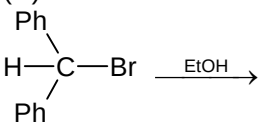
(i) The order of rate of E_2 reaction is



(ii) The poorest leaving group among the – SH, – NH_2, – OAc, – OH is – OH.

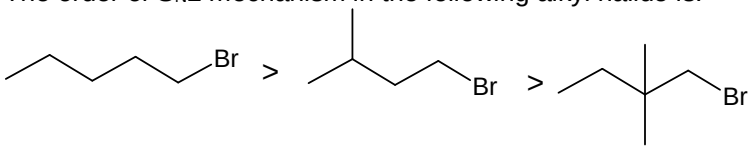
(iii)  is stronger nucleophile as compared to Et_3N.

(iv) In the reaction



Rate of reaction is unaffected, if the concentration of Ph_2CHBr is tripled and the concentration of ethanol is doubled.

(v) The order of S_N2 mechanism in the following alkyl halide is:



Space for Rough work

SECTION – D

(Numerical Answer Type)

This section contains **3 questions**. Each question, when worked out will result in **numerical answer type** (XXXXX.XX)

44. There are five solutions
Solution 1 – 0.2 M HCl
Solution 2 – 0.2 M BOH ($K_b = 10^{-2}M$)
Solution 3 – 0.2 M, 1 lit HCl + 0.2 M, 1 lit BOH
Solution 4 – 0.2 M, $K_4[Fe(CN)_6]$
Solution 5 – 0.2 M, Na_3PO_4
What is the correct code of solution numbers in increasing order of their respective freezing points?

45. At 298 K, the emf of the cell
$$Hg \,|\, 0.01\,mol\,dm^{-3}\ kcl \,|\, 1\,mol\,dm^{-3}\,KNO_3 \,|\, 0.01\,mol\,dm^{-3}\ KOH \,|\, Hg$$
(saturated with Hg_2Cl_2; saturated with HgO)
is found to be 0.1634 v. What is the free energy change of the cell reaction? (Given 1F = 96500 C)

46. The density of a sodium chloride sample is found to be 2.16801 gm/cm^3, instead of 2.18 gm/cm^3. What is the percentage of unoccupied sites in the sample?

Space for Rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains **3 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

47. A circle C_1 with radius 5 touches x-axis and another circle C_2 with radius 4 touches y-axis. The two circles touches each other externally so that their point of contact lies in the first quadrant, let the locus of their point of contact be the curve S. Two tangents are drawn to the curve S from point P(9, 5) to meet the curve at Q and R. The area of ΔPQR is

(A) $\frac{29}{4}$ (B) $\frac{27}{5}$

(C) $\frac{13}{2}$ (D) $\frac{25}{3}$

48. In a ΔABC, internal angle bisectors AI, BI and CI are produced to meet opposite sides in A′, B′ and C′ respectively, then the maximum value of $\frac{AI \cdot BI \cdot CI}{AA' \cdot BB' \cdot CC'}$ is

(A) $\frac{27}{2}$ (B) $\frac{13}{4}$

(C) $\frac{8}{27}$ (D) $\frac{27}{256}$

49. Point 'O' is the centre of ellipse with major axis AB and minor axis CD. Point F is one focus of ellipse. If OF = 6 and the diameter of the inscribed circle of ΔOCF is 2, then the product AB.CD equals

(A) 65 (B) 72

(C) 78 (D) 92

Space for rough work

Multiple Correct Choice Type

This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

50. Vertices A, B and C of a tetrahedron ABCD are (1, 1, 1), (1, 0, 0), (3, 0, 0) respectively. The altitude from vertex D to the opposite face ABC meets the median line through A of the ΔABC at a point E. If the length of side AD is 4 and volume of the tetrahedron is $\frac{2\sqrt{2}}{3}$, then the correct statements is/are

(A) the altitude from vertex D is 2
(B) there is exactly one position for the point E
(C) there can be two position for the point E
(D) vector $\hat{j}-\hat{k}$ is normal to the plane ABC

51. Let $\alpha = e^{\frac{i\pi}{2017}}$, then

(A) $\prod_{r=1}^{2016}\left(\alpha^r+\alpha^{-r}\right)=2$
(B) $\text{Re}\left(\sum_{r=1}^{2016}\alpha^r\right)=0$
(C) $\sum_{r=1}^{2016}\left(\alpha^r+\alpha^{-r}\right)=0$
(D) $\sum_{r=1}^{2016}\left(\alpha^r+\alpha^{-r}\right)=1$

52. Let $P\left(\alpha,\frac{1}{\alpha}\right)$, $(0 < \alpha < 1)$ be any point on the hyperbola xy = 1. Tangent to the hyperbola at P intersect the coordinate axes at Q and R & normal at P intersects the coordinate axes at M and N (Q and M lie on x-axis). If ΔOQR and ΔOMN have equal area, then

(A) area of $\Delta OMN = 2$
(B) $\alpha=\sqrt{2}-1$
(C) $QR = 2^{7/4}$
(D) $MN = 2^{7/4}$

53. The roots of equation $x^5 - 40x^4 + \alpha x^3 + \beta x^2 + \gamma x + \delta = 0$ are real and in G.P. If the sum of their reciprocal is 10, then δ can be

(A) –32
(B) $\frac{1}{32}$
(C) $-\frac{1}{32}$
(D) 32

Space for rough work

54. A function y = f(x) satisfying the differential equation $\frac{dy}{dx}\cdot\sin x - y\cos x + \frac{\sin^2 x}{x^2} = 0$ is such that $y \to 0$ as $x \to \infty$, then

(A) $\lim_{x\to 0} f(x) = 1$

(B) $\int_0^{\pi/2} f(x)dx$ is less than $\frac{\pi}{2}$

(C) $\int_0^{\pi/2} f(x)dx$ is greater than unity

(D) f(x) is an odd function

Comprehension Type

This section contains **1 group of questions.** The group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct

Paragraph for Question Nos. 55 to 56

Read the following write up carefully and answer the following questions:
Consider the following three sets of complex roots of unity. A = {z: $z^{18} = 1$}, B = {ω: $\omega^{48} = 1$} and C = {zω: z ∈ A and ω ∈ B}, then

55. The number of distinct elements in the set C is

(A) 864

(B) 48

(C) 66

(D) 144

56. Let the set C has n distinct elements $C_1, C_2, ..., C_n$, then the value of $|C_2 - C_1| + |C_3 - C_1| + ... + |C_n - C_1|$ is

(A) $2\cot\left(\frac{5}{8}\right)^\circ$

(B) $\cot\left(\frac{5}{4}\right)^\circ$

(C) $2\cot\left(\frac{15}{11}\right)^\circ$

(D) $\cot\left(\frac{15}{8}\right)^\circ$

Space for rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **10 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

57. f: R $\to$ R is a continuous function satisfying $\int_0^1 \left(\frac{f(x)}{x}\right)^2 dx = 2\int_0^1 f(x)dx - \frac{1}{3}$, then $256f\left(\frac{1}{8}\right)$ is equal to _____

58. There are seven letters L_1, L_2, L_3, L_4, L_5, L_6 and L_7 and six envelopes E_1, E_2, E_3, E_4, E_5, E_6 corresponding to first six letters. The total number of ways if none of the letters goes to its correct envelope, even letters goes to even envelopes. Odd letters goes to envelopes other that E_4 and E_6 and all the envelopes except E_2 contains exactly one letter is, N. If $N = 2\alpha + 4$, where $\alpha \in z$, then α is equal to _____

59. Let $A(z_1)$, $B(z_2)$ and $C\left(z_1 + \frac{z_2 - z_1}{1-i}\right)$ are vertices of a triangle. If $|z_2 + iz_1| = |z_1| + |z_2|$, $|z_1| = \sqrt{7}$ and $|z_2| = 3$, then area of ΔABC is _____

60. Let $A = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + ... + \frac{1}{\sqrt{10000}}$, then find $\left[\frac{[A]}{50}\right]$ _____ (where [.] denotes the greatest integer function)

61. The possible number of ordered triplets (m, n, p) where m, n, $p \in N$ is (6250)k. Such that $1 \le m \le 100$, $1 \le n \le 50$, $1 \le p \le 25$ and $2^m + 2^n + 2^p$ is divisible by 3, then k is _____

62. Number of integral values of p for which the system of equations $\cos^{-1}x + \left(\sin^{-1}y\right)^2 = \frac{p\pi^2}{4}$ and $\left(\cos^{-1}x\right)\left(\sin^{-1}y\right)^2 = \frac{\pi^4}{16}$ has real solutions is _____

Space for rough work

63. Let f: $R^+ \to R^+$ be a differentiable function satisfying $f(x)\cdot f(yf(x)) = f(x + y)\ \forall\ x, y \in R^+$ and $f'(0) = -1$, and $f(0) = 1$ then area of the region bounded by the curve $y = \frac{f(x)}{x+1}$ and the coordinate axes is _____

64. Let $I = \int_0^{\ln 2} \frac{2e^{3x} + e^{2x} - 1}{e^{3x} + e^{2x} - e^x + 1} dx$, then $11e^{-I}$ is equal to _____

65. The number of different hyperbolas, which can be represented by the equation $1 = \sqrt{x^2 + y^2} - {}^nC_r x$ is _____ (where n and r are integers satisfying $1 \le r \le n \le 5$)

66. Let $A = \begin{bmatrix} 1 & \frac{3}{2} \\ 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 4 & -3 \\ -2 & 2 \end{bmatrix}$ and $C_r = \begin{bmatrix} r\cdot 3^r & 2^r \\ 0 & (r-1)3^r \end{bmatrix}$ be given matrices. If $\sum_{r=1}^{50} t_r\left((AB)^r C_r\right) = 3 + a\cdot 3^b$ _____ (where $t_r(A)$ denotes trace of matrix A), then value of $\frac{a+b}{20}$ is (where $a, b \in I$ and a is not divisible by 3)

SECTION – D

(Numerical Answer Type)

This section contains **3 questions**. Each question, when worked out will result in **numerical answer type** (XXXXX.XX)

67. The equation $a_8x^8 + a_7x^7 + a_6x^6 + ... + a_0 = 0$ has all roots positive and real and $a_8 = 1$, $a_7 = -4$ and $a_0 = \frac{1}{2^8}$, then the value of $\frac{a_2a_6}{a_4}$ is equal to _____

68. Coordinate of any point P are (4, 5). If A and B are variable points on straight lines y = x and y = 2x respectively. Then find minimum value of $(PA + PB + AB)^2$ _____

69. If the length of the largest interval in which the function $f(x) = \sin^{-1}|\sin \alpha x| + \cos^{-1}(\cos \alpha x)$ is constant is 2π, then the value of α is _____ where ($\alpha > 0$)

Space for rough work